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Q1 Let P be an EC point. What is the minimum number of EC sums/doubles necessary to compute $[259]P$?

- ☐ a) 8
- ☒ b) 10
- ☐ c) 11
- ☐ d) 12
- ☐ e) 258
- ☐ f) 259

Q2 What is the main limitation of a trivial secret sharing scheme?

- ☐ a) Unlike the Shamir scheme, it is not ideal
- ☐ b) Unlike the Shamir scheme, it is not unconditionally secure but only computationally secure
- ☐ c) It permits only to implement (t,n) schemes with t strictly lower than n
- ☒ d) It permits only to implement (n,n) schemes and not (t,n) schemes with $t < n$

Q3 In the Boneh-Franklin's Identity Based Encryption scheme, what happens if an attacker compromises the PKG?

- ☐ a) Nothing, as there is no PKG in such scheme
- ☐ b) It becomes impossible to decrypt a previously encrypted data
- ☒ c) the attacker may find all private keys for all users
- ☐ d) the attacker may revoke all users' public keys

Q4 Three parties A, B, C setup a group $(3,3)$ RSA signature, i.e. a message is correctly signed if all three parties contribute to the signature with their shares of the private key d . Being x and y random values (in the appropriate range), shares are:

Share_A = $d-x-y$

Share_B = x

Share_C = y

Assuming that a message M needs to be signed, schematically describe the specific modular operations and exchange of messages that such a $(3,3)$ RSA signature requires.

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Q5 What may happen if Alice digitally signs two different messages M1 and M2, with ECDSA using the same nonce r ($r = x\text{-coordinate}(kP) \bmod n$)?

- ☒ **a)** The attacker can compute Alice's Private key
- ☐ **b)** The attacker can forge a signature for any linear combination of M1 and M2
- ☐ **c)** The attacker can decrypt both M1 and M2
- ☐ **d)** The attacker can perform an expansion attack on one of the two messages

Q6 Assume arithmetic modulus 100. A Linear secret sharing scheme involving 4 parties is described by the following access control matrix:

A:	1	1	1
B:	0	1	0
C:	0	0	1
D:	0	0	-1

Assume that the following shares are revealed:

A $\rightarrow$ 36
B $\rightarrow$ 51
D $\rightarrow$ 18

What is the secret?

- a)** 3 **b)** 5 **c)** 31 **d)** 33 **e)** 67 **f)** 69 **g)** 95 **h)** 97 **i)** another result = _____

Q7 Describe the threshold El Gamal decryption, and specifically explain why the private key is never revealed in the reconstruction.

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Q8 A same message M is RSA-encrypted using two different public keys $e_1 = 11$ and $e_2 = 17$, but same RSA modulus $n=35$. The two resulting ciphertexts are: $c_1=3$ and $c_2=17$. Decrypt the message applying the Common Modulus Attack (show the detailed computations required).

[Just in case you might need to rapidly compute inverses mod 35, see table associated to exercise Q10]

Answer: by the extended GCD(17,11) $\rightarrow \{r,s\}=\{2,-3\}$

Hence

$$M = 3^{-3} \times 17^2 \bmod 35 = 12^3 \times 17^2 \bmod 35 = 12$$

Q9 Consider the Elliptic curve $y^2 = x^3 + 2x - 1$ defined over the modular integer field $\mathbb{Z}_5$. A) find all the points $EC(\mathbb{Z}_5)$ and B) specify what is the order of the corresponding group

O, {0,2}, {0,3}, {2,1},{2,4},{4,1},{4,4}

Order 7

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Q10 A Shamir Secret Sharing scheme uses a non-prime modulus $p=35$ (if you need modular inverses see table on the right). Of the 5 participating parties $P_1, \dots, P_5$, with respective x coordinates $x_i = \{1, 2, 3, 4, 5\}$, parties P_1 , P_2 and P_5 aim at reconstructing the secret.

a) compute the Lagrange Interpolation coefficients for parties 1, 2, 5;

b) Reconstruct the secret, assuming that the shares are:

$P_1 \rightarrow 18$

$P_2 \rightarrow 24$

$P_5 \rightarrow 19$

c) Prove that the system is NOT unconditionally secure, by showing that the knowledge of the two shares P_1 and P_5 leak information about the secret – specifically, after knowing shares P_1 and P_5 which would be the only possible remaining secret values?

[Answer: Secret = 14;

set of possible secrets: the 7 possible values which satisfy $19+10x \bmod 35 \rightarrow$

$\rightarrow \{4, 9, 14, 19, 24, 29, 34\}$

x	1/x mod 35
1	1
2	18
3	12
4	9
6	6
8	22
9	4
11	16
12	3
13	27
16	11
17	33
18	2
19	24
22	8
23	32
24	19
26	31
27	13
29	29
31	26
32	23
33	17
34	34